

✓ Phase 2 - Security & Privacy

```
import pandas as pd
df = pd.read_csv('/content/cleaned_data.csv')
```

Caesar Cipher encryption function

```
def caesar_cipher_encrypt(text, shift=3):
    result = ''
    for char in text:
        if char.isalpha():
            base = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char) - base + shift) % 26 + base)
        else:
            result += char
    return result
```

the text columns to encrypt

```
text_columns = ['city', 'region', 'type']
```

encryption on text columns

```
for col in text_columns:
    df[col] = df[col].astype(str).apply(caesar_cipher_encrypt)
```

Save the encrypted dataset

```
df.to_csv('encrypted_data.csv', index=False)
```

Access Control

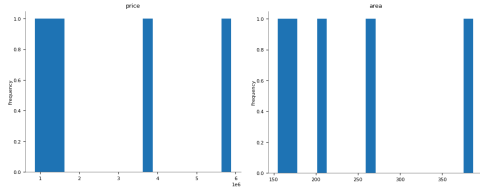
```
def access_data(role):
    if role == 'admin':
        return df
    elif role == 'user':
        return df[['price', 'area']]
    else:
        return "Access Denied"
```

Admin access

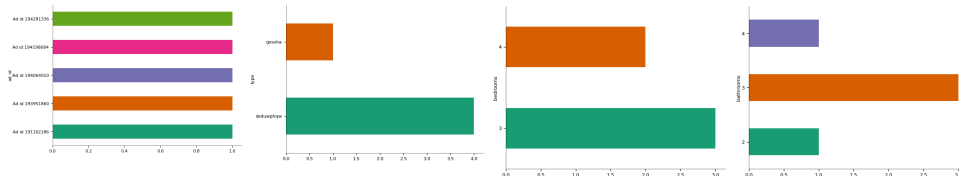
```
admin_view = access_data('admin')
admin_view.head()
```

	ad_id	type	price	area	bedrooms	bathrooms	level	furnished	rent	city	region
0	Ad id 194281336	dsduwphqw	3790000.0	207.0	3	3	2	NaN	no	jlcd	6wk ri rfwrehu
1	Ad id 194198684	dsduwphqw	870000.0	155.0	3	3	4	no	no	fdlur	qhz kholrsrolv
2	Ad id 191102186	dsduwphqw	1546400.0	170.0	3	2	9	no	no	fdlur	cdkudd do pddgl
3	Ad id 193951860	dsduwphqw	1300000.0	270.0	4	3	ground	no	no	vkdutld	10wk ri udpdgdq
4	Ad id 194064010	gxsoha	5888000.0	387.0	4	4	ground	no	no	fdlur	qhz fdlur - ho wdjdpd

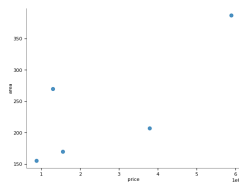
Distributions



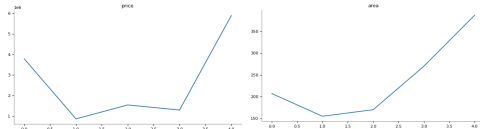
Categorical distributions



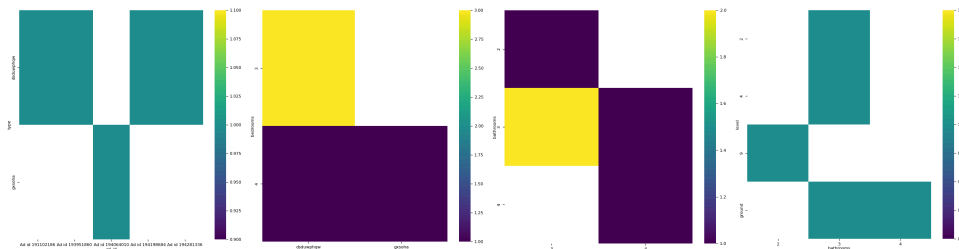
2-d distributions



Values



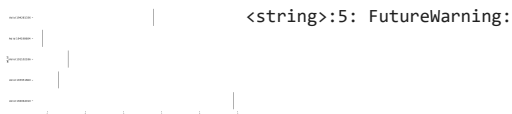
2-d categorical distributions



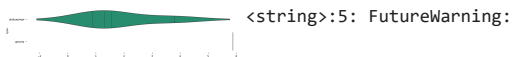
Faceted distributions

<string>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend`



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Next steps:

User access

Generate code with admin_view

☒ View recommended plots

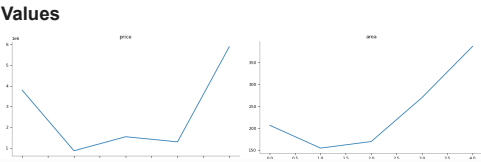
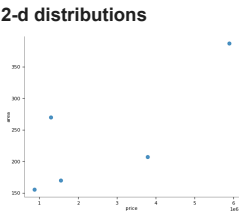
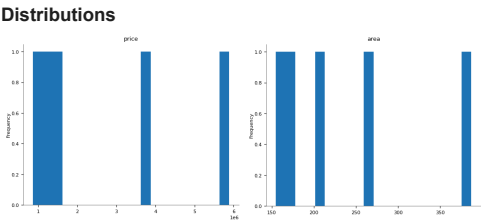
New interactive sheet

```
user_view = access_data('user')
user_view.head()
```



	price	area
0	3790000.0	207.0
1	870000.0	155.0
2	1546400.0	170.0
3	1300000.0	270.0
4	5888000.0	387.0





Next steps:

Generate code with user_view

☒ View recommended plots

New interactive sheet

Test Guest access

```
denied_view = access_data('guest')
print(denied_view)
```

 Access Denied